

HURC - Hatchling Universal Robot Controller

Hatchling Controller

Compatible Controllers:

- HURC dev2

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Class Documentation

1 NetController Class Reference

Public Member Functions

- **NetController** (bool displayDebugMessage=false)
 - Creates a "network controller" - basically a receiver for your HURC. Make sure to call `controllerSetup()` in your `setup()` function before attempting to receive inputs.
- void **controllerSetup** ()
 - sets up stuff related to the controller, put this in ur setup function
- float **getJoy1X** ()
 - Returns the left joystick's x value
- float **getJoy1Y** ()
 - Returns the left joystick's y value
- bool **getJoy1Z** ()
 - Returns whether or not the left joystick is clicked down
- float **getJoy2X** ()
 - Returns the right joystick's x value
- float **getJoy2Y** ()
 - Returns the right joystick's y value
- bool **getJoy2Z** ()
 - Returns whether or not the right joystick is clicked down
- bool **getA** ()
 - Returns whether or not the A button is pressed down (Xbox layout)
- bool **getB** ()
 - Returns whether or not the B button is pressed down (Xbox layout)
- bool **getX** ()
 - Returns whether or not the X button is pressed down (Xbox layout)
- bool **getY** ()
 - Returns whether or not the Y button is pressed down (Xbox layout)
- bool **getLb** ()
 - Returns whether or not the left bumper is pressed down
- bool **getRb** ()
 - Returns whether or not the right bumper is pressed down
- bool **getLt** ()
 - Returns whether or not the left trigger is pressed down
- bool **getRt** ()
 - Returns whether or not the right trigger is pressed down
- bool **getDpad** ()
 - Returns whether or not the dpad button is pressed down

2 Constructor & Destructor

2.1 NetController()

```
NetController::NetController (  
    bool displayDebugMessage = false)
```

Creates a "network controller" - basically a receiver for your HURC. Make sure to call `controllerSetup()` in your `setup()` function before attempting to receive inputs.

Parameters

<code>displayDebugMessage</code>	Default is false; If set to true, it will print out debug messages to the serial monitor. if enabled, make sure to do <code>Serial.begin()</code> before you initialize your <code>NetController</code> . This will flood your monitor, so only enable it when necessary.
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3 Member Functions

3.1 getJoy1X()

```
float NetController::getJoy1X ()
```

Returns a float of the left joystick's x value in the range [-1, 1]

3.2 getJoy1Y()

```
float NetController::getJoy1Y ()
```

Returns a float of the left joystick's y value in the range [-1, 1]

3.3 getJoy1Z()

```
bool NetController::getJoy1Z ()
```

Returns true when the left joystick is held down, false otherwise

3.4 getJoy2X()

```
float NetController::getJoy2X ()
```

Returns a float of the right joystick's x value in the range [-1, 1]

3.5 getJoy2Y()

```
float NetController::getJoy2Y ()
```

Returns a float of the right joystick's y value in the range [-1, 1]

3.6 getJoy2Z()

```
bool NetController::getJoy2Z ()
```

Returns true when the right joystick is held down, false otherwise

3.7 getA()

```
bool NetController::getA ()
```

Returns true when the A button is held down, false otherwise (Xbox layout)

3.8 getB()

```
bool NetController::getB ()
```

Returns true when the B button is held down, false otherwise (Xbox layout)

3.9 getX()

```
bool NetController::getX ()
```

Returns true when the X button is held down, false otherwise (Xbox layout)

3.10 getY()

```
bool NetController::getY ()
```

Returns true when the Y button is held down, false otherwise (Xbox layout)

3.11 getLb()

```
bool NetController::getLb ()
```

Returns whether or not the left bumper is held down, false otherwise

3.12 getRb()

```
bool NetController::getRb ()
```

Returns whether or not the right bumper is held down, false otherwise

3.13 getLt()

```
bool NetController::getLt ()
```

Returns whether or not the left trigger is held down, false otherwise

3.14 getRt()

```
bool NetController::getRt ()
```

Returns whether or not the right trigger is held down, false otherwise

3.15 getDpad()

```
bool NetController::getDpad ()
```

Returns whether or not the dpad button is held down, false otherwise

Non-Class Documentation

1 Reference

- bool `printMacAddress` ()
 - *Prints the ESP32 Mac Address in the serial monitor.*

2 Functions

2.1 `printMacAddress()`

`bool NetController::getRt ()`

Prints the ESP32 Mac Address in the serial monitor. Recommended to run in `setup()`. Make sure to run `Serial.begin(115200)` and update `platformio.ini` with “`monitor_speed = 115200`” before calling.